

Milestone 2 Progress Report

Project: DeckForge – New-Player Onboarding (Pokémon TCG Archetype Browser)

Course: IAT 461 – Assignment 4

Date: August 6, 2026

Current Stage

Both project phases are complete in `eda.ipynb`.

Phase 1: EDA & Strategy Validation

- Loaded and inspected `pokemon_cards.csv` (816 cards, 14 columns).
- Audited missing values by supertype; confirmed all gaps are structural (Trainer/Energy cards lack HP, attacks, and abilities by design), not data errors.
- Evaluated all three proposed features against the data:
 - Option B** (price tracker): eliminated — no price column and no reliable proxy for market value.
 - Option C** (power-level warning): eliminated — no ground-truth power label exists in the dataset.
 - Option A** (archetype browser): **selected** — all needed signal (attack stats, ability flags, effect text) is present in `attacks_json` and `abilities_json`.

Phase 2: Data Science Implementation

- Feature engineering stage 1: parsed nested JSON columns into numeric features (attack count, damage, energy cost, damage-per-energy, retreat count, evolution stage flags).
- Feature engineering stage 2: mined attack/ability text for 10 binary mechanic flags (draw, search, heal, status, energy acceleration, discard, coin flip, etc.).
- Clustered 624 Pokémon cards with K-Means on the scaled feature matrix; swept $k = 2$ to 8 using silhouette score and selected $k = 6$.
- Named and interpreted all 6 clusters using centroid profiles and example cards (e.g. Basic Attackers, Bulky Heavy Hitters, Ability Disruptors, Evolved Mid-Range, Energy Accelerators, Agile Strikers).
- Handled Trainer (153) and Energy (39) cards separately by subtype, since they have no attack/ability data and would collapse into a meaningless cluster otherwise.
- Wrote conclusions and recommendations section summarizing both phases for the client.

What’s Left

- Polish visualizations (PCA cluster plot, centroid heatmap) for presentation quality.
- Light proofreading pass on markdown commentary throughout the notebook.
- Final review against the client’s success criteria before submission.

Challenges / Notes

- Silhouette scores across $k = 2$ to 8 were modest overall (best at $k = 6$, ~ 0.184), expected for card-game mechanic data since archetypes naturally overlap rather than form tight, separable clusters.
- Trainer/Energy cards required a separate, rule-based grouping approach since clustering does not apply to them.